

Student's name

Teacher's name

2019

10 5.2 Course Semester 2 Examination

General Instructions

- Reading time – 2 minutes
- Working time –90 minutes
- Write using blue or black pen
Black pen is preferred
- Scientific calculators may be used

Total Marks – 60

- Questions arranged in topics. Attempt questions from all topics.

Section	1	2	3	4	Total
	$\overline{24}$	$\overline{12}$	$\overline{36}$	$\overline{23}$	

Student Comment:

Teacher Comment:

PRACTICE PAPER

Linear and Non Linear Relationships

24 marks

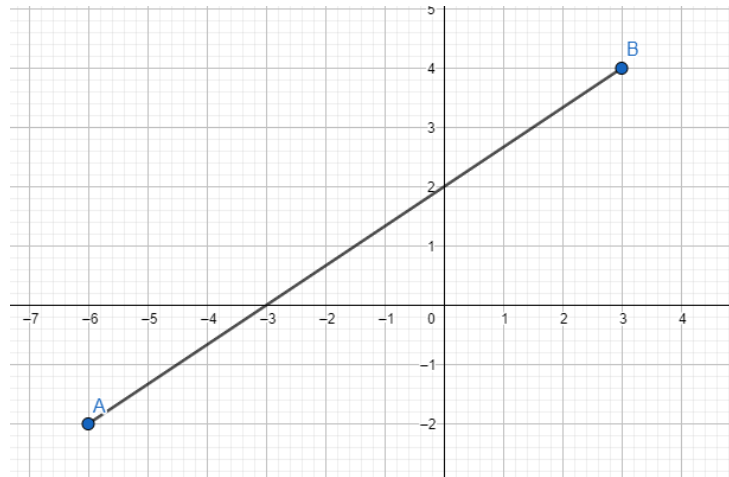
Attempt all Questions

Questions 1 to 4 Multiple choice

Allow about 15 - 20 minutes for this section

Circle the correct answer for Questions 1 to 5

Questions 1-2 refer to the diagram below



1. The gradient of interval AB is:

(A) $\frac{9}{6}$

(B) 2

(C) $\frac{2}{3}$

(D) -3

D4

2. The mid-point of interval AB is:

(A) (0, 2)

(B) (-1.5, 1)

(C) (4.5, 3)

(D) (-3, 0)

D4

3. Line EF has a gradient of $-\frac{3}{2}$. Line CD is perpendicular to line EF. The gradient of line CD is:

(A) $-\frac{3}{2}$

(B) $-\frac{2}{3}$

(C) $\frac{2}{3}$

(D) $\frac{3}{2}$

C5

4. Which one of the following lines does the point (2, -2) lie on?

C6 (A) $y = -3x + 4$

(B) $y = 4x + 9$

(C) $y = 3x + 2$

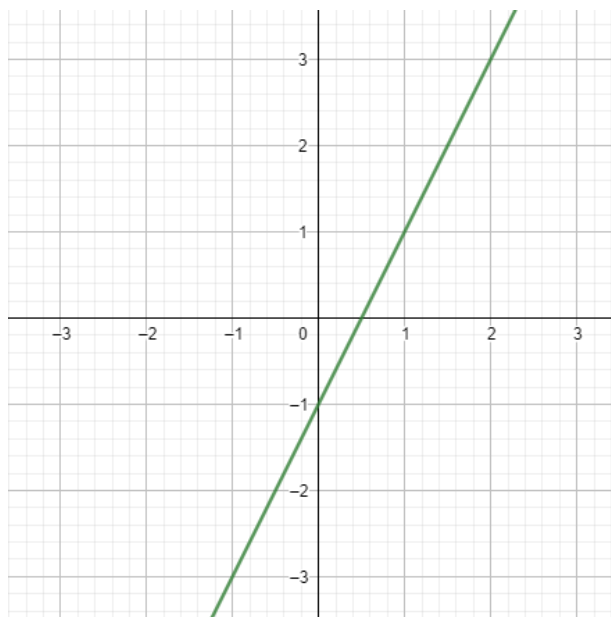
(D) $y = -2x$

Questions 5-8 Short Answer. Show all working. Marks are as shown

5. Parts a, b, and c refer to the graph shown below.

Marks

C5



a. Determine the gradient of the line

1

C5

$$m = 2$$

b. Find the y-intercept of the line

1

C5

$$b = -1$$

c. Write the equation of the line in the form $y = mx + b$

2

C5

$$y = 2x - 1$$

6.

Marks

a. A line has the equation $y = 4x - 6$. What is the gradient of a line parallel to this line?

1

C5

$$m = 4$$

- b. Write the equation of the line that has the same y-intercept as the line $y = 2x + 3$, and is parallel to the line $y = -\frac{x}{4} - 4$ 2

C6

$$y = -\frac{x}{4} + 3$$

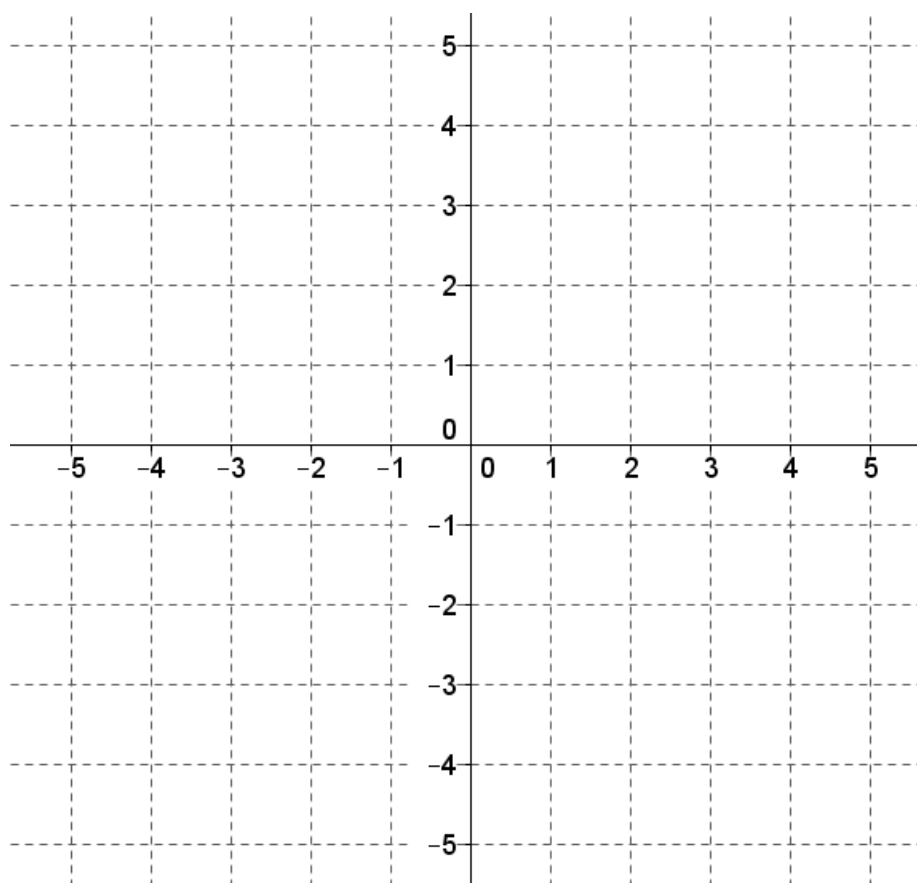
- c. What is the equation of a line perpendicular to the line $y = 2x + 3$ that goes through the point $(2, 5)$ 3

C6

$$y = -\frac{1}{2}x + 6$$

7. Sketch the graphs given in parts a to c in the axes below

Marks



- a. $x = -4$ 1

D4

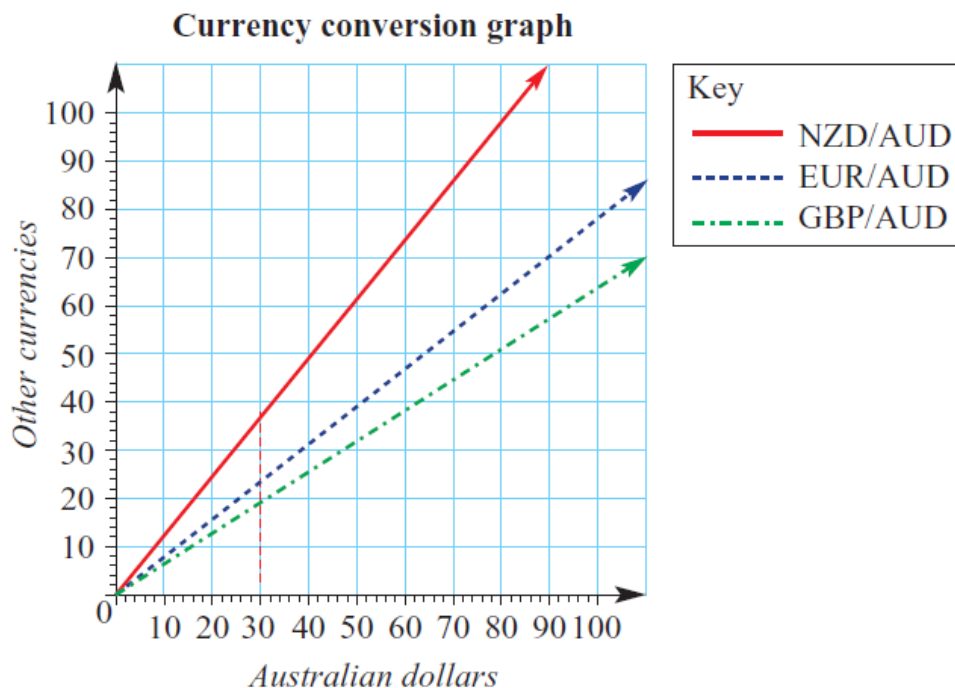
- b. $y = 4 - x$ 2

D4

- c. $y = 3x - 4$ 2

D4

8. Use the length unit conversion graph to answer questions.



2

- a. Convert 70 AUD to NZD and use these values to find the gradient of the line, to 2 decimal places.

C5

86 ←

$$\frac{86}{70} = 1.23$$

1

- b. Use your previous answer to find the constant of proportionality k and write the direct proportion equation between NZD (y) and AUD (x).

B7

$$y = 1.23x$$

2

- c. Use the equation to convert 86 NZD to AUD.

C6

$$x = \$69.92$$

Questions 10 to 14 Multiple choice

Right- Angled Triangles

12 marks

Attempt all Questions

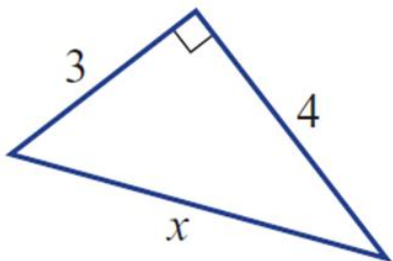
Questions 9 to 12 Multiple choice

Allow about 15-20 minutes for this section

Circle the correct answer for Questions 10 to 13

9. The correct working out to calculate the side x in the triangle shown below is:

E2



(A) $x = \sqrt{(4^2 - 3^2)}$

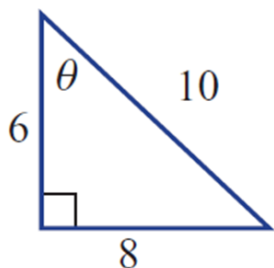
(B) $x = \sqrt{(4^2 + 3^2)}$

(C) $x = 4 - 3$

(D) $x = 4 + 3$

10. In the triangle below the ratio $\tan\theta$ is given by:

D3



(A) $\frac{8}{10}$

(B) $\frac{8}{6}$

(C) $\frac{6}{10}$

(D) $\frac{8}{10}$

11. A measurement of an angle is found to be $82^\circ 40' 15''$. What is this measurement rounded to the nearest degree?

D4

(A) 82°

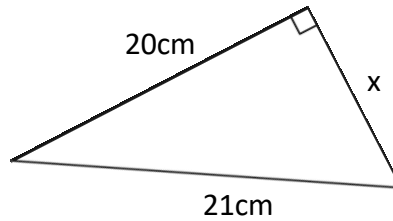
(B) 83°

(C) 43°

(D) 44°

12. What is the length of the side marked x ?

D4



(A) 41cm

(B) 6.4m

(C) 29cm

(D) 6.4cm

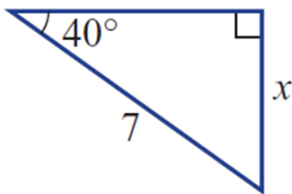
$x =$ (D) 6.4cm

Questions 13-14 Short Answer. Show all working. Marks are as shown

13. Find the side length x in these right-angled triangles.

Marks
2

a.

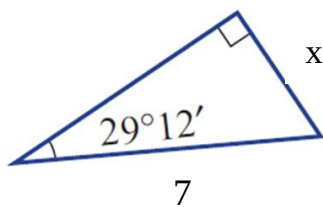


D4

$$x = 4.5$$

b.

2



D4

$$x = 3.42$$

14

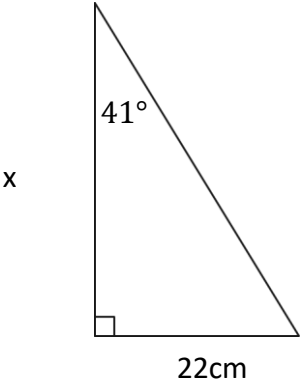
Find the missing angle θ in these right-angled triangles, to the nearest minute

Marks

a.

2

D4

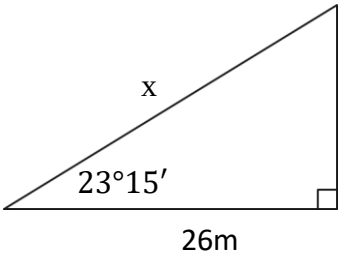


25.31

b.

2

D4



$x = 28.30$

Probability

36 marks

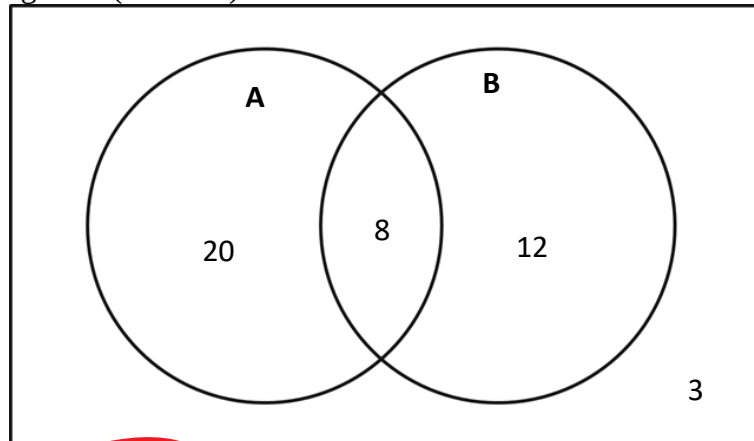
Attempt all Questions

Questions 15 to 18 Multiple choice

Allow about 15- 20 minutes for this section

Circle the correct answer for Questions 15 to 18

15. In the following Venn Diagram $P(A \text{ and } B)$ is:



D3

- (A) $3/43$ (B) $8/43$ (C) $32/43$ (D) $40/43$

16. In the following two-way table, find the number of elements in 'not A'.

	A	not A	
B	7	12	19
not B	13	6	19
	20	18	38

D3

- (A) 6 (B) 12 (C) 18 (D) 38

17. An experiment involves tossing three coins and counting the number of heads.

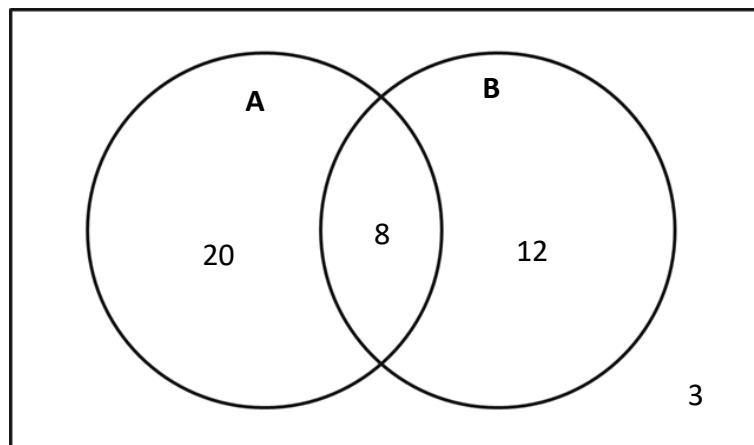
Number of heads	0	1	2	3
Frequency	6	26	19	9

D4

What is the experimental probability of at least 2 heads?

- (A) 19 (B) 28 (C) $19/60$ (D) $28/60$

18. In the following Venn Diagram $P(A | B)$ (the probability of event A occurring given that event B has occurred) is:



B7

(A) $8/43$

(B) $20/43$

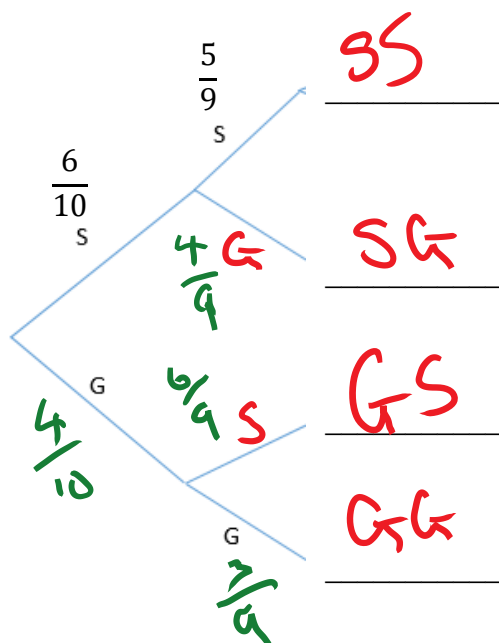
(C) $8/20$

(D) $8/28$

Questions 19-23 Short Answer. Show all working. Marks are as shown

19. A bag contains 6 silver coins and 4 gold coins. ~~Two~~ ^{Two} coins are chosen without replacement.

Marks



- a. Complete the tree diagram above and list the sample space.

C6

4

$\{SS, SG, GS, GG\}$

- b. What's the probability of obtaining at least 1 silver coin?

C6

3

$13/15$

- c. What's the probability of obtaining exactly 1 gold coin?

C6

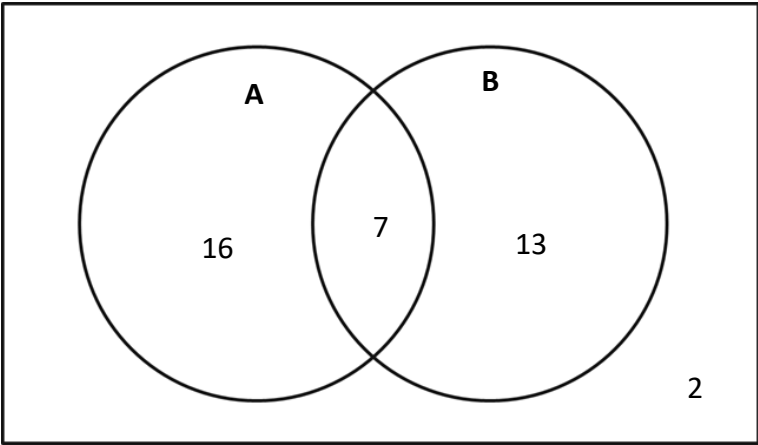
2

$8/15$

20.

The Venn diagram shows the distribution of elements in two sets, A and B.

Marks



a.

Transfer the information in the Venn diagram to a two-way table

9

D4

	A	not A	
B	7	13	20
not B	16	2	18
	23	15	38

D3

D4

b.

Find the number of elements in A or B.

2

36

c.

Find P(not A).

2

15/38

21.

A bakery sells 3 flavour doughnuts: plain (P), strawberry (S), and chocolate (C). Steph is in a hurry and randomly selects 2 doughnuts.

Marks

a.

Complete this table showing the possible combinations of doughnuts selected

2

C6

	P	S	C
P	(P,P)	(S,P)	(C,P)
S	(P,S)	(S,S)	(C,S)
C	(P,C)	(S,C)	(C,C)

C6

b.

Find the probability that Steph selects a strawberry doughnut.

2

5/9

22 Given that a number greater than 3 has been rolled on a fair six-sided die:

a List the sample space.

1

B7

$\{4, 5, 6\}$

b Calculate the probability that this number was 6.

2

B7

$\frac{1}{3}$

c Calculate the probability that this number was 2.

1

B7

0

23 “If you toss a coin and obtain a head three times in a row, there is a greater chance the fourth toss will result in a tail.”

2

Is this statement correct or incorrect? Explain your answer.

B7

Incorrect.

(Eg. Independent events)

Statistics

23 marks

Attempt all Questions

Questions 24 to 27 Multiple choice

Allow about 15- 20 minutes for this section

Circle the correct answer for Questions 24 to 27

24. What type of data would the following question generate?

“How would you rate the final episode of Game of Thrones? Excellent, Neutral, Awful.”

E2

(A) Categorical –
Nominal

(B) Awful

(C) Numerical – Discrete

(D) Categorical –
Ordinal

25. The results of a mathematics exam are given below. Calculate the mean.

34 82 56 72 91 96 65 76 72 61 58 72 81 42 62

D3

(A) 62

(B) 68

(C) 72

(D) 76

26. What is the median of the following set of data?

19 10 12 8 6 3 14 13

D3

(A) 10

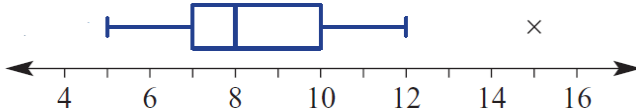
(B) 11

(C) 12

(D) 7

27. What is the outlier, as represented in this box plot?

C5



(A) 5

(B) 8

(C) 12

(D) 15

Questions 28-30 Short Answer. Show all working. Marks are as shown

28. The amount of points scored each game by 2 AFL teams are recorded below:

Marks

GWS	107	65	59	83	56	106	67	101	98	88	72	77
Swans	109	95	56	81	76	51	62	66	82	72	69	66

a. Calculate the mean, median and range for each team.

6

D3

GWS
mean 81.58
median 80
range 51

Swans
73.75
70.5
58

- D4** b. Which team achieved the better scores overall and justify your answer. 2
Use your results from part (a).

GWS higher mean
and median

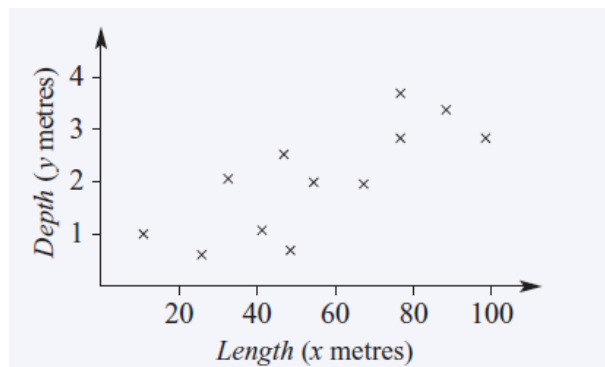
- D4** c. Represent this data in a back-to-back stem-and-leaf plot. 3

GWS	S	Swans
9 6	5	1 6
7 5	6	2 6 6 9
7 2	7	2 6
8 3	8	1 2
6	9	5
7 6 1	10	9

5 | 1 means
51

- 29.** Describe the correlation between the depth and length of 11 pools in the area. Marks

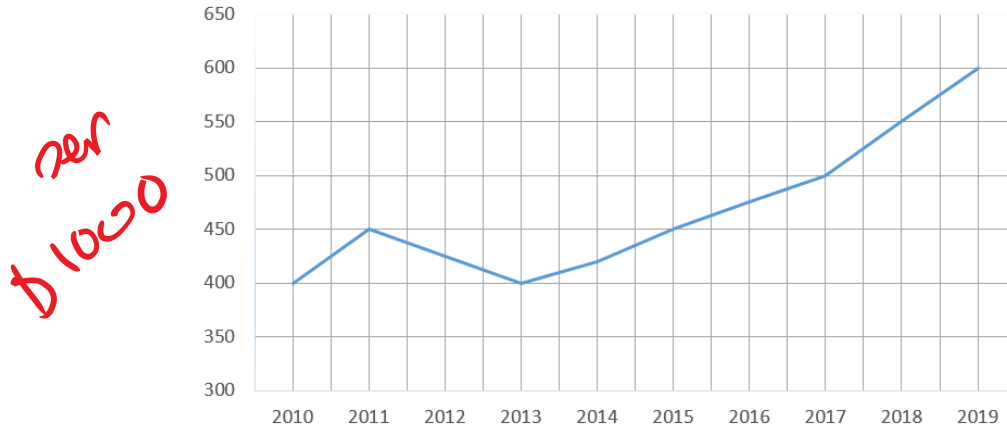
- C6** 2



weak, positive

30. This time-series plot shows the trend of Australian house prices over a 10 year period.

Marks



- a Describe the general trend in house prices over this time period. 2
C6 increases 2010 - 2011 decrease until 2013
continuous increase since 2013
- b When was the average house price at its lowest during this time period? 2
C6 2010 and 2013 when it was
\$400,000
- c What was the average house price in 2016? 2
C6 \$475,000 in 2016

END OF EXAM

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